Accuracy Is Not Reliance: Estimating How Much Medical Vision-Language Models Use the Image, with Time-Matched Interventions

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Abstract

A model can score well on a medical benchmark without looking at the evidence in the image. I wanted a number that says how much a model's answer depends on that evidence, not how often the answer is right. I use pairs of chest X-rays from the same patient, taken at different times, where a finding (for example pleural effusion) appears or disappears. The change in the model's output on such a pair is compared with the change on a second pair from the same patient, at the same time gap, where the finding does not change. I call the difference R_{dir}. In a simulation where the true answer is known, a model that ignores the evidence is wrongly flagged as reliant in 100% of runs without any control and in 2% of runs with my time-matched control. On the NIH ChestX-ray14 dataset, MedGemma-4B shows clear reliance on effusion evidence (R_{dir} = 0.16), while Qwen2.5-VL-7B shows none on four findings and three prompt wordings. When I train a classifier with a planted shortcut, its benchmark AUC rises from 0.82 to 1.00 while its reliance on the real evidence falls from 0.076 to 0.000. When I add a fake prior report to the prompt, an “effusion present” report makes MedGemma answer “yes” to every image.

1. Introduction

Medical vision-language models (VLMs) are usually judged by accuracy. That tells us how often they are right. It does not tell us why. Several recent studies found that medical VLMs often answer well even when the image is removed or replaced [1, 2, 3]. So a high score can hide a model that leans on text, priors, or shortcuts.

My question is narrow: *when the evidence for a finding changes in the image, does the model's answer change with it?* I call this evidence reliance. It is different from accuracy. A model can be accurate and not reliant, if it uses a shortcut that happens to agree with the label. It can also be reliant and not very accurate.

The obvious way to test this is to edit the image, but edits make images that look unlike real X-rays. I avoid editing. I use real follow-up X-rays of the same patient. The catch is that two X-rays taken days apart differ in many ways: position, exposure, and the patient's overall condition. A model can react to those changes too. My design tries to remove that part.

What I did:

- I defined a directional, time-matched reliance estimand, R_{dir}, and a sign-based version, R_{sign}, with a patient-level bootstrap for confidence intervals (Section 3).
- I checked both in a simulation where the true reliance is known (Section 5).
- I measured three models on four findings in NIH ChestX-ray14 (Section 6).

- I ran two controlled experiments where the right answer is known in advance: a planted image shortcut and a fake prior report in the prompt (Section 7).

2. Related work

Visual grounding of medical VLMs. Zafar et al. [1, 2] compare accuracy with real, blank, shuffled and swapped images on medical VQA benchmarks. Lotfinia et al. [3] run a causal audit of chest X-ray VLMs and report that they do not always need the image. Xiong et al. [4] study visual attribution for chest X-ray reasoning. Moll et al. [5] test whether chain-of-thought explanations stay faithful under image and text perturbations. Most of these works measure differences in accuracy. I measure the change in the model's output on the same patient, and I do not need the model to be correct.

Image editing for stress tests. RadEdit [6] and RoentMod [7] edit chest X-rays with generative models to find shortcuts. I use real follow-up images instead, which keeps the images realistic but brings in time drift. That is the reason for my time-matched control.

Statistics of variable importance. My estimand is related to conditional permutation importance [8], knockoffs and the conditional randomization test [9], the holdout randomization test [10], and model reliance [11]. Those methods swap a feature for a draw that carries *no* information about the label. My intervention does the opposite: the flip partner carries *different* information about the label, and the control partner carries the *same*.

Shortcuts in chest X-rays. Models are known to use chest drains as a cue for pneumothorax [12]. My planted-marker experiment copies this situation with a marker I control.

3. Method

3.1 Pairs from the same patient

For a finding (say effusion), I pick an **anchor** image a of a patient. I then take:

- a **flip partner** f: another image of the same patient and view, where the label is the opposite, at follow-up gap k from the anchor;
- a **control pair** (s_1, s): two images of the same patient and view with the *same* label, and *exactly the same signed gap* k. If several pairs qualify, I prefer the anchor's label and then the pair closest in time to the anchor.

Each patient gives at most one anchor. This keeps patients independent.

3.2 Estimands

Let $p(\cdot)$ be the model's probability that the finding is present, and $\Delta y = y_f - y_a$, which is $+1$ if the finding appears and -1 if it disappears.

$$D_{flip} = E[(p(f) - p(a)) \cdot \Delta y], \quad D_{ctrl} = E[(p(s) - p(s_1)) \cdot \Delta y], \quad R_{dir} = D_{flip} - D_{ctrl}$$

$\mathrm{D_{flip}}$ asks: does the score move toward the new label? $\mathrm{D_{ctrl}}$ asks the same question on a pair where the label did not change, so any movement there comes from time drift. The subtraction removes drift that is shared by both pairs. A model that ignores the evidence has $\mathrm{R_{dir}}$ near 0.

R_{sign} is the same formula with sign(·) applied to each difference. It only counts the direction of movement. It does not change if the model's scores are rescaled by any increasing function.

Labels are used only to set the direction of the change. I never score the model's answer as right or wrong.

3.3 Validity checks

- **Same time gap.** The control pair must have the same signed gap as the flip pair (Section 5 shows why).
- **A real yes/no answer.** For VLMs, P(yes) is computed from the next-token probabilities of “yes” and “no”. I keep only anchors where these two tokens hold at least 90% of the next-token probability on all four images. Section 6.3 shows that this matters.
- **Other cues held fixed.** Any marker or text cue must be the same on all four images. Section 7 shows what happens otherwise.

3.4 Inference

I use a patient-level bootstrap (2,000 resamples) for 95% percentile intervals, a one-sided sign test on per-patient R_{dir}, and paired bootstraps on shared patients for model comparisons. When I compare many pairs of models I adjust p-values with the Benjamini–Hochberg procedure.

4. Data and models

Data. NIH ChestX-ray14 [13] has 112,120 frontal X-rays from 30,805 patients, with labels mined from radiology reports by text processing. Each image has a follow-up number, so I can order a patient's images in time. I use effusion as the main finding, and atelectasis, pneumothorax and cardiomegaly as extra findings. For effusion I get 1,445 balanced anchors (750 where the finding appears, 695 where it disappears).

Models.

- A DenseNet-121 trained on CheXpert [14], from TorchXRayVision [15]. It was not trained on NIH, so it is a fair positive control. I also fine-tune it on NIH in Section 7.
- Qwen2.5-VL-3B and Qwen2.5-VL-7B [16], general-purpose VLMs.
- MedGemma-4B [17], a medical VLM.

The main prompt is “Is there a pleural effusion in this chest X-ray? Answer yes or no.” All runs used a Kaggle notebook with T4 GPUs. Qwen2.5-VL-7B took about 0.9 s per image in float16, and MedGemma about 3.8 s per image in float32.

5. Simulation study: does the estimator find the truth?

On real data I cannot know the true reliance. So I simulated patients where I can. Each patient has four images. The model's log-odds are $\beta_e \cdot y + \beta_z \cdot z$ + noise, where y is the label and z is a nuisance that drifts over time. The drift is correlated with the direction of the label change. I did this on purpose: on NIH, the control pairs drift in the same direction as the label change ($D_{ctrl} > 0$, Section 6). The true reliance is computed by

changing only y and holding z fixed. Each setting has 300 repeats with 1,000 bootstrap draws.

Estimator (model ignores the evidence, n = 400)	**Mean estimate**	**False positives (nominal 2.5%)**
No control (D_{flip} only)	0.087	100%
Control at a different time gap, R_{dir}	0.039	47%
Control at a different time gap, R_{sign}	0.096	28%
Control at the same time gap, R_{dir}	0.003	**2.0%**
Control at the same time gap, R_{sign}	0.002	**2.0%**

Table 1. Simulation with a model that does not use the evidence at all. Without a control, or with a control at a different time gap, drift is mistaken for reliance.

What else I found:

- **R_{dir} works as an effect size.** Its 95% intervals covered the truth in 91–97% of runs when the effect was zero, small or moderate. It has a small negative bias (about −4% of the true value), and coverage drops for a strong effect (0.90 at n = 400, 0.76 at n = 1,000). My guess is that the control removes drift only approximately because probabilities are not linear.
- **R_{sign} is a good test but a poor effect size.** It keeps false positives at 2% under the null, but its size is pulled down by nuisance noise (−16% to −20% relative bias for larger effects). So I use R_{dir} for size and R_{sign} for testing and for comparing models with different calibration.
- **Label noise shrinks the estimate.** If 20% of observed flips are fake, R_{dir} falls to 77% of its true value. If 40% are fake, it falls to 58%. So estimates on noisy labels are lower bounds.
- **Saturation.** Pushing scores toward 1 with the same ranking drives the true R_{dir} to about 0 (0.196 → 0.003), while R_{sign} stays the same.

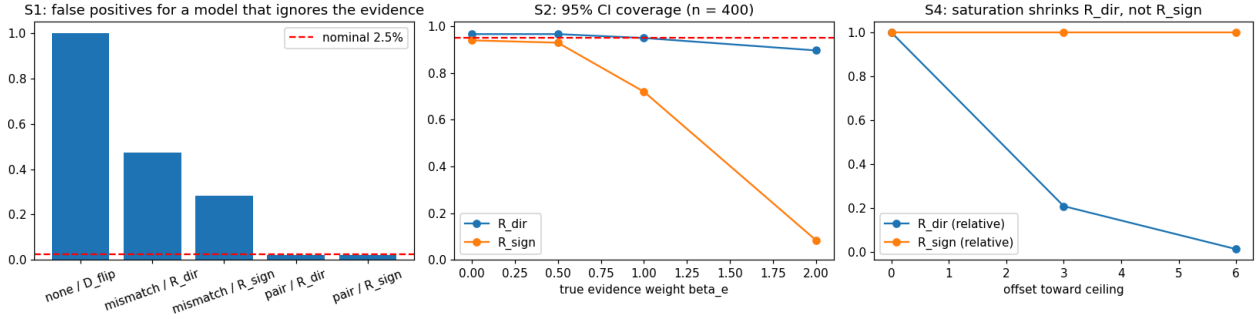


Figure 1. Simulation. Left: false-positive rate for a model that ignores the evidence. Middle: coverage of 95% intervals (n = 400). Right: as scores approach the ceiling, R_{dir} shrinks but R_{sign} does not.

6. Results on NIH ChestX-ray14

6.1 The control matters on real data

In the first real run (400 effusion anchors), the CheXpert DenseNet had D_{flip} = 0.095 and D_{ctrl} = 0.043 [0.016, 0.069]. So the same-label control pairs moved toward the new label too. Without the control, I would have reported about 80% more reliance (0.095 instead of 0.052). I also tried an absolute version, |p(f) − p(a)| minus |p(s) − $p(s_1)$|. It looked "significant" for Qwen2.5-VL-7B on the log-odds scale (0.089) even though Qwen did not move toward the new label at all. That is why the directional version is my main estimand.

6.2 The labels are noisy

The DenseNet's AUC for effusion was 0.768 [0.739, 0.798] on 1,000 random NIH images but only about 0.62 on the images in flip pairs. Pairs where the text-mined label changes seem to contain more label errors or borderline cases. Together with the simulation (Section 5), this means my real-data numbers are probably underestimates.

6.3 Three models, four findings, three prompts

Finding	DenseNet AUC (random images)	DenseNet R_{dir}	Qwen2.5-VL-7B R_{dir}	MedGemma-4B R_{dir}	MedGemma, yes/no mass ≥ 0.9
Effusion	0.77	0.063 [0.041, 0.082]	0.001 [−0.006, 0.008]	0.160 [0.094, 0.225]	0.167
Atelectasis	0.65	0.047 [0.028, 0.065]	−0.009 [−0.033, 0.014]	0.061 [0.026, 0.099]	**0.000**
Pneumothorax	0.58	0.016 [−0.000, 0.030]	−0.002 [−0.004, 0.001]	0.059 [−0.004, 0.122]	0.060
Cardiomegaly	0.62	0.028 [−0.001, 0.054]	0.006 [−0.025, 0.038]	0.114 [0.050, 0.183]	0.046 [−0.006, 0.103]

Table 2. R_{dir} with 95% patient-bootstrap intervals. DenseNet uses all anchors (740–1,445 patients); the VLMs use 400 balanced anchors per finding (298 for cardiomegaly).

Four things stand out.

1. **Qwen2.5-VL-7B shows no reliance anywhere.** It was near zero on all four findings and on three wordings of the effusion question, including a negated one ("Is this chest X-ray free of a pleural effusion?"). The differences between wordings were 0.000 and 0.010, both with intervals covering 0. Qwen2.5-VL-3B was also null in an earlier run (−0.001 [−0.011, 0.010]).
2. **MedGemma relies on effusion evidence.** The result held under all three wordings (0.160, 0.182, 0.090), under the yes/no filter, and under R_{sign} (0.26–0.27). It also beat Qwen on R_{dir} for effusion (paired difference 0.159 [0.091, 0.226]).
3. **Some apparent reliance was fake.** For atelectasis, MedGemma's R_{dir} was 0.061. But on anchors where it gave a clean yes/no answer, it said "yes" to every image and R_{dir} became 0.000. The signal came only from prompts where the model was not

really answering yes or no. This is why the yes/no check is a required step, not an extra.

4. **The DenseNet is a good positive control only for effusion.** Its AUC on the other findings was 0.58–0.65, so those findings cannot show whether the method works.

One more caution. MedGemma's scores are extreme: 94.6% of its P(yes) values were below 0.01 or above 0.99. MedGemma beat the DenseNet on R_{dir} for effusion (−0.081 [−0.146, −0.016] for DenseNet minus MedGemma) but not on R_{sign} (−0.013 [−0.160, 0.143]). So I only claim that MedGemma is at least as reliant as the DenseNet on effusion, not more.

7. Controlled experiments

7.1 A planted image shortcut

Here I know the answer in advance. I fine-tuned the CheXpert DenseNet with a new effusion output on 8,000 NIH images from patients that were never used for testing. On each training image I stamped a small checkerboard in the top-left corner. It appeared on a fraction ρ of positive images and 1 − ρ of negative images. At ρ = 0.5 the marker is useless. At ρ = 1.0 it predicts the label perfectly. I trained two seeds for each ρ and tested on the 1,445 effusion anchors and 1,000 held-out random patients.

ρ	AUC, marker as in training	AUC, no marker	R_{dir}, marker fixed	R_{dir}, marker follows label	Marker effect
0.5	0.82	0.82	0.076 [0.058, 0.095]	0.076	0.006
0.7	0.85	0.81	0.074	0.14	0.16
0.9	0.95	0.79	0.043 [0.031, 0.057]	0.53	0.61
1.0	**1.00**	0.70	**0.000 [−0.004, 0.004]**	0.95	0.95

Table 3. Seed 0 intervals shown; the two seeds agreed within 0.002 on R_{dir}. “Marker effect” is the average change in P(effusion) when the marker is added to the same image.

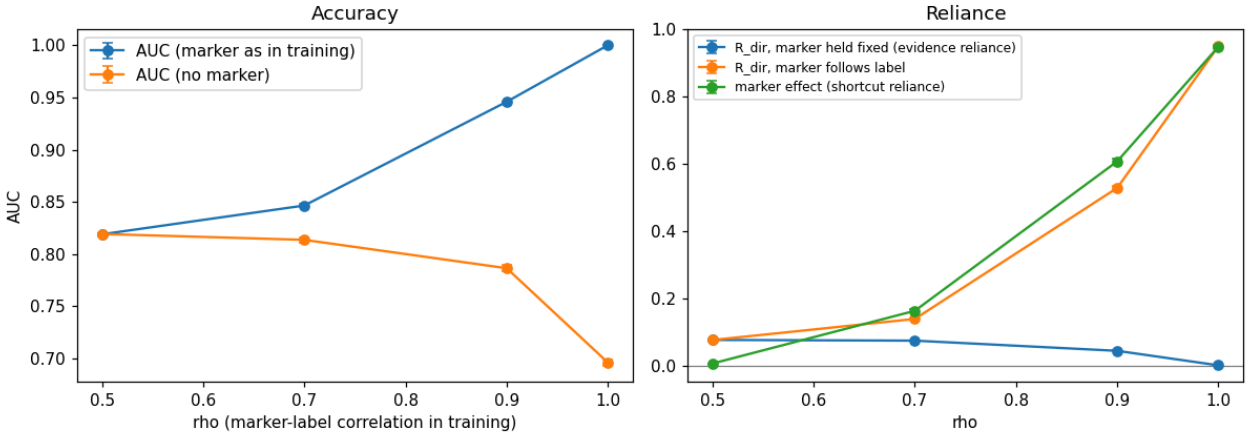


Figure 2. As the shortcut gets stronger, the AUC a benchmark would report goes up, while reliance on the real evidence goes down. If the marker is allowed to follow the label, R_{dir} tracks the shortcut instead of the evidence.

This is the main result of the project. Accuracy and reliance move in opposite directions, and R_{dir} follows the reliance. It also shows why the cue must be held fixed: when the marker follows the label, R_{dir} reaches 0.95, which only measures the marker.

A caveat. At ρ = 1.0 the model still reaches AUC 0.70 without the marker, so it has not fully stopped using the evidence. R_{dir} reads 0 partly because, without the marker, its scores sit near 0 (median 0.012). On the log-odds scale, reliance falls from 0.41 to 0.13, and R_{sign} falls from 0.20 to 0.11. So reliance drops on every scale, but "zero" holds only on the probability scale.

7.2 A fake prior report in the prompt

I added one sentence before the question: "Prior radiology report: pleural effusion is present." or "Prior radiology report: no pleural effusion." I used the same 400 effusion patients and kept only anchors with yes/no mass ≥ 0.9.

Condition	Qwen "yes" rate	Qwen R_{dir}	MedGemma "yes" rate	MedGemma R_{dir}	MedGemma R_{sign}
No report	0.00	0.001	0.62	0.167 [0.101, 0.237]	0.28
Report "present" (all images)	0.38	0.001	**1.00**	**0.000**	0.28
Report "absent" (all images)	0.00	0.000	0.29	0.162 [0.093, 0.231]	0.28
Truthful report (matches label)	0.19	0.440	0.60	0.773	0.97
Text effect: P(yes | present) − P(yes | absent)	0.437 [0.424, 0.451]		0.684 [0.645, 0.721]		

Table 4. Qwen = Qwen2.5-VL-7B; MedGemma = MedGemma-4B. MedGemma rows have 312–381 patients after the yes/no filter; Qwen rows have 400.

- **Qwen follows the text and ignores the image.** The report moves its answer a lot (0.44). Its image reliance stays at about 0 in every condition.
- **MedGemma uses both, but a positive report wins.** With "present" it says "yes" to every image. With "absent" the image can still change its answer (R_{dir} 0.162, almost the same as with no report).
- **The image still moves MedGemma's score under a positive report.** R_{dir} is 0 but R_{sign} stays at 0.28. The score changes, but never enough to change the answer. The gap between R_{dir} and R_{sign} is a useful sign that another cue is overriding the image.
- **A truthful report makes a model look reliant.** For Qwen, R_{dir} jumps from 0.001 to 0.440 although it uses nothing from the image. Measures that let the text change with the label mix up text and image.

8. Limitations

- **Noisy labels.** NIH labels come from text mining. My simulation and the AUC gap (0.77 vs 0.62) suggest the real numbers are underestimates. I have not yet checked

them on radiologist-verified pairs.

- **Few models and one dataset.** Two VLM families, one CNN, one public dataset. I have not tested larger models.
- **The scale matters.** R_{dir} on the probability scale measures the effect on the answer. It shrinks near 0 or 1 even if the ranking is unchanged (Sections 5 and 7). My results should be read with this in mind.
- **Small bias for strong effects.** In simulation, R_{dir} undercovered for strong effects at large n.
- **No formal theory.** I state the assumptions (no drift that differs between the flip pair and the control pair at the same gap, and a well-posed yes/no answer) but I have not proved identification.
- **A yes/no format only.** Free-text answers would need a different scoring rule.

9. Conclusion

Accuracy and reliance on evidence are different things, and they can move in opposite directions. A simple design with real follow-up X-rays and a control pair at the same time gap can measure reliance without editing images and without grading answers. In my tests, it caught a planted shortcut that pushed benchmark AUC to 1.00. It showed that Qwen2.5-VL-7B does not use the image for these questions. It also showed that one sentence of fake history makes MedGemma ignore the image when the history says the finding is present.

Reproducibility and acknowledgements

All experiments ran in Kaggle notebooks with public data (NIH ChestX-ray14) and public model weights. The simulation uses fixed seeds; re-running it on Kaggle gave identical numbers. Code, notebooks and result files are in the repository linked above. I used an AI assistant (Claude, by Anthropic) for discussing the study design, writing code, and drafting this report. I ran all experiments, checked the results against the output files, and take responsibility for every claim.

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